



Ameliorative effect of silymarin against linezolid-induced hepatotoxicity in methicillin-resistant *Staphylococcus aureus* (MRSA) infected Wistar rats



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ABSTRACT

Linezolid has a better choice for eradication of methicillin-resistant *Staphylococcus aureus* (MRSA) infections, but its use is limited because of linezolid-induced hepatotoxicity, myelosuppression, and lactic acidosis. This research elucidated the role of silymarin against hepatotoxicity of linezolid therapy in MRSA infected Wistar rats. The rats were rendered neutropenic by an intraperitoneal injection of cyclophosphamide injection. The neutropenic rats were injected subcutaneously with 10^6 CFU/ml of MRSA. The rats were divided into 6 groups. Normal control, Infected, Infected animals treated with linezolid 50 mg/kg/twice/day and Infected animals treated with linezolid and different dose of silymarin 25, 50, and 100 mg/kg/twice/day for 14 days. On the 15th day, the blood, liver, kidney, and bone marrow were collected for biochemical and histopathological examination. The MRSA was confirmed by PCR assay. The minimum inhibitory concentration of linezolid was 0.5–2 µg/ml. The linezolid induced liver damage was confirmed by elevation of marker enzymes alanine aminotransferase (ALT), aspartate aminotransferase (AST), alkaline phosphatase (ALP), Lactate dehydrogenase (LDH) levels, serum bilirubin, lactate, and histopathological studies of the liver. The linezolid treated rats also showed myelosuppression, lactic acidosis, oxidative stress and decreased intestinal alkaline phosphatase (IAP). The silymarin administration exhibited marked hepatoprotective effect by significantly lowering the liver marker enzymes, serum parameters, and cytological findings reflect the hepatoprotection. Additionally, Silymarin showed protection against myelosuppression and lactic acidosis evidenced by bone marrow smear and serum lactate estimation. Antioxidant effect of silymarin was confirmed by decreased levels of lipid peroxidation, restored the enzymatic and non-enzymatic antioxidants of the liver nearer to normal. The present study indicates that the silymarin could be a better herbal therapeutic agent which protects against the linezolid induced hepatotoxicity in MRSA infected rats.

1. Introduction

Methicillin-resistant *Staphylococcus aureus* (MRSA) is responsible for both hospital and community-acquired bacterial infections [1]. MRSA is resistant to several types of antibiotics. It causes life-threatening infections because it is difficult to treat or become worse if fair treatment is delayed. In the hospital, MRSA cause surgical infections, catheter infections and invasive infections such as soft tissue infections, bone infections, and bloodstream infections [2]. Furthermore, currently available antibiotic to treat MRSA infection is limited. The Linezolid (LZD) is oxazolidinone antibiotic superior over vancomycin in the eradication of MRSA [3]. LZD has a unique structure and mechanism of action, which targets protein synthesis at an exceedingly early stage. LZD has 100% bioavailability, when given orally and allows conversion to oral therapy when the patient is clinically stable. It reduces the

length of hospital stay, cost, intravenous infections and increased patient convenience [4]. Consequently, LZD has no cross-resistance with other commercially available antimicrobial agents. Linezolid-resistant *Staphylococcus aureus* isolates are very rare, > 99% of isolates being susceptible with linezolid in surveys [5].

LZD is a relatively safe antibiotic when given for short periods. Prolong LZD therapy cause hepatotoxicity, lactic acidosis; peripheral and optic neuropathy, and bone marrow suppression, but diabetic soft tissue infections, bone infections, and chronic infections required to prolong LZD therapy. Silymarin (SN) is flavonolignan, obtained from seeds of *Silybum marianum* L. It is an antioxidant [6], hepatoprotective [7], antidiabetic [8], anti-inflammatory [9], antifibrotic [10], and cytoprotective [11]. SN has been used medicinally to cure liver diseases including viral hepatitis, cirrhosis and alcoholic liver disorders. Hepatoprotective mechanism of SN is due to increased hepatocyte

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regeneration, increase in the reduced glutathione level in the liver and decrease the binding of hepatotoxin to receptor sites on the hepatocyte [12]. The study was designed to assess the protective role of silymarin against linezolid induced hepatotoxicity in MRSA infected rats.

2. Materials and methods

2.1. Drugs and chemicals

Linezolid and Silymarin were obtained from Sigma-Aldrich Chemical Pvt Limited, India. Bovine serum albumin, Thiobarbituric acid, Trichloroacetic acid, butylated hydroxyl toluene, oxidized glutathione, epinephrine was obtained from Himedia laboratories, Mumbai, India. All other drugs and chemicals used in the study were obtained commercially and were of analytical grade. For estimation of biochemical tests, kits were obtained from Ecoline, Manufactured by Merck Specialties, Private limited, Ambernath.

2.2. Experimental animals

The Wistar rats (200–250 g) were maintained in clean, sterile, polypropylene cages with paddy husk as bedding. Animals were housed at a temperature of $25 \pm 2^\circ\text{C}$ and humidity of 30–60%. A 12:12 h light and the dark cycle were followed. The animals had free access to food and water ad libitum. All the experimental procedures and protocols used in this study were reviewed by the Institutional animal ethics committee (688/2/CPCSEA) and were in accordance with the guidelines of the IAEC. Animal care was given as per the guidelines of Committee for the Purpose of Control and Supervision of Experiments on Animals (CPCSEA).

2.3. Microorganism

The clinical isolate of MRSA with positive in both Pantone-Valentine leukocidin (Pvl) toxin and methicillin resistance (*mecA*) gene was obtained from the ARM laboratory, Erode, Tamil Nadu, India.

2.4. PCR screening

A duplex-PCR was used to screen the presence of *mecA* and *lukS/F-PV* (which encode the PVL S/F bicomponent proteins) genes [13]. For rapid DNA extraction, 2–5 colonies were suspended in 100 μl of molecular-grade water (Qiagen, Germany) and heated in a boiling water bath for 10 min. After centrifugation at 10,000 rpm for 10 min, 1 μl of the supernatant (DNA template) was added to 20 μl of the PCR reaction mixture (Invitrogen, USA). The reaction was carried out using the following conditions: an activation step at 94°C for 5 min, followed by 30 cycles of initial-denaturation at 94°C for 30 s, 50°C for 30 s, and 72°C for 30 s, ending with a final extension step at 72°C for 7 min, and followed by a holding step at 4°C . The PCR products were analyzed by gel electrophoresis with 2 percent agarose in Tris/Borate/EDTA buffer with ethidium bromide (5 $\mu\text{g}/\text{mL}$) and visualized by UV-transillumination. A 100 bp DNA ladder (Invitrogen, USA) was used as a marker Duplex-PCR yielded the products with sizes of 433 and 310 bp which corresponded to *lukS/F-PV* and *mecA* genes respectively.

2.5. Determination of MIC using broth microdilution assays

Microbroth dilution assays were performed according to the Clinical and Laboratory Standards Institute guideline (CLSI) [14]. A single colony was transferred and incubated overnight at 37°C with Mueller Hinton broth. The diluted culture was measured for the absorbance at OD_{620} to get a concentration of 1×10^6 CFU/ml. A total of 100 of the freshly diluted cultures and serially diluted antibiotics were dispensed into 96-well culture plates. The plates were incubated at 37°C for 16–18 h. The last well in the series without any visible growth was read as MIC.

2.6. Experimental design

Rats were rendered neutropenic by intraperitoneal cyclophosphamide injections given for 4 days and 5th day at a dose of 150 mg/kg and 100 mg/kg, respectively. This neutropenia was maintained for 5 days. The neutropenic rats were injected subcutaneously with 0.2 ml of the MRSA suspension containing approximately 10^6 CFU/ml in 5% hog gastric mucin [15]. Animals received treatment of anti-infective compounds starting 2 h post infection. LZD was administered orally at a dose of 50 mg/kg twice daily for 14 days [16]. SN was administered orally at a dose of 25, 50, and 100 mg/kg twice daily for 14 days [17].

The animals are divided into 6 groups ($n = 10$)

Group I: Control

Group II: Rats were infected with MRSA

Group III: Infected rats treated with LZD 50 mg/kg twice daily

Group IV: Infected rats treated with LZD & SN 25 mg/kg twice daily

Group V: Infected rats treated with LZD & SN 50 mg/kg twice daily

Group VI: Infected rats treated with LZD & SN 100 mg/kg twice daily

On 15th-day rats, abscess size was traced on 1 mm² graph paper. Rats were anesthetized and blood samples were collected into EDTA added vials for hematological study and into dry non-heparinised tubes for serum biochemical profile, and the serum was separated by centrifuging at 3000 rpm for 15 min. The abscesses were excised, weighed and added to 2 ml of heart infusion broth, and homogenized. The number of viable organisms in the abscess (the number of CFU per abscess) was determined by a standard plate procedure with mannitol salt agar. The liver and kidney tissue portions were dissected and then fixed in 10% formalin for histopathological studies. Sections of 5–6 μ thickness were prepared and stained by hematoxylin and eosin staining agents. Bone marrow aspirations of femur bone were collected and their smears prepared immediately and stained for bone marrow examination.

2.7. Preparation of intestinal homogenate

The intestine was removed, flushed with saline and washed again. The mucosa was scraped off with the help of a microscopic glass slide. 10% W/V of the mucosa was mixed with 5 mM EDTA (pH-7.4) and homogenized with Teflon pestle [18]. The homogenate was centrifuged and the supernatant liquid was used for estimation of IAP.

2.8. Preparation of liver homogenate

The liver was minced and homogenized in 0.05 M ice-cold phosphate buffer pH 7.4 by using a mechanical blender to make 10% homogenate. The homogenate was used for estimation of the protein [19], Malondialdehyde and lipid peroxidation [20]. The remaining part of the homogenate was mixed with 10% trichloroacetic acid in the ratio of 1:1, centrifuged at 5000 g for 10 min at 4°C and supernatant was used for estimation of Superoxide dismutase (SOD) [21], Catalase (CAT) [22], Glutathione reductase (GSSH) [23], Glutathione peroxidase (GPx) [24], Peroxidase (Px) [25], Reduced glutathione (GSH) [26], Vitamin A [27], Vitamin C [28], and Vitamin E [29].

2.9. Statistical analysis

The data are expressed as the mean $\pm$ standard deviation (SD). GraphPad Prism software was used to analyze data and construct the graphs. One-way analysis of variance (ANOVA) followed by Dunnett's test. The values of $p < 0.05$ were regarded as statistically significant.

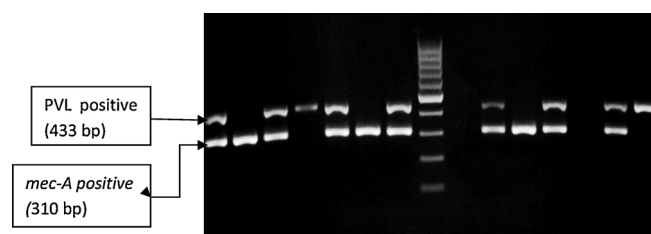


Fig. 1. Duplex – PCR molecular characterization for PVL and *mec A* genes in MRSA.

3. Results

3.1. *In vitro* antimicrobial activity

The clinical isolate of MRSA was confirmed by duplex PCR (Fig. 1). The clinical and laboratory standards institute clinical breakpoints were considered for interpretation of LZD MIC (sensitive $\leq 4 \mu\text{g/ml}$ and resistant $\geq 8 \mu\text{g/ml}$) *S. aureus* ATCC 29213 was used as a control strain for MIC detection. MRSA isolates tested, was sensitive to LZD ranged from $0.5 \mu\text{g/ml}$ – $2 \mu\text{g/ml}$. LZD showed well *in vitro* activity against MRSA isolates and can be considered for the treatment of these infections.

3.2. Abscess size and Bacterial count

The abscess size of infected animals showed 189.2 mm^2 (Table 1). The LZD treated infected animals showed a significant reduction in abscess size (72.4 mm^2) compared to infected rats. The SN 25, 50 and 100 mg/kg and LZD treated infected animals showed significant ($p < 0.01$) reduction in the size of abscess (64.6 , 31.5 , and 28.2 mm^2) when compared to infected animals. The number of viable organisms in the abscess (number of CFU per abscess showed in Table 1) was measured by serial dilution colony counting method. The infected animals showed 4.20×10^6 CFU/abscess (Fig. 2a and b). The LZD treated infected animals showed significant ($p < 0.01$) reduction in the bacterial count (7.22×10^3 CFU/abscess) than infected animals. The SN 25, 50 and 100 mg/kg and LZD treated infected animals showed significant ($p < 0.01$) decrease in the bacterial count (3.26 , 1.71 , and 1.12×10^3 CFU/abscess) when compared to infected animals. The abscess size was correlated with the bacterial count.

3.3. Hematological parameters

We found a significant reduction ($p < 0.01$) in RBC, WBC, PLT, and Hb in LZD treated infected animals (Table 2) when compared to normal animals. The SN 50, and 100 mg/kg restored the normal RBC and mild elevation of the WBC, PLT and Hb levels than LZD treated group.

Table 1

Effect of silymarin on abscess size and MRSA count in infected and LZD treated MRSA infected rats.

Group	Abscess size in mm^2	Abscess weight in mg	MRSA CFU/ Abscess
Infected (I)	189.2 ± 8.6	34.2 ± 1.6	$4.20 \times 10^6 \pm 0.02$
I + LZD	72.4 ± 6.8^a	18.1 ± 1.2^a	$7.22 \times 10^3 \pm 0.01^a$
I + LZD + SN (25)	64.6 ± 6.2^a	16.9 ± 1.4^a	$3.26 \times 10^3 \pm 0.02^a$
I + LZD + SN (50)	31.5 ± 5.6^a	15.4 ± 0.9^a	$1.71 \times 10^3 \pm 0.04^a$
I + LZD + SN (100)	28.2 ± 4.2^a	14.9 ± 1.1^a	$1.12 \times 10^3 \pm 0.03^a$

Values are mean $\pm$ S.D; n = 10 in each group. $^aP < 0.01$, when compared to infected animals (one way ANOVA followed by Dunnett's test). Data were analyzed by one way ANOVA followed by Dunnett's test.

^a Significant difference in comparison with the infected group ($P < 0.01$).

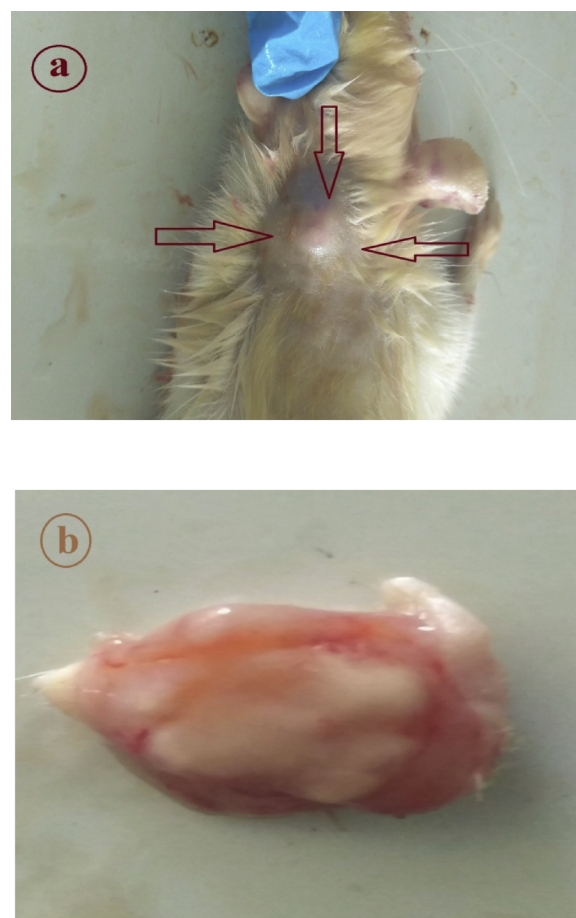


Fig. 2. a & b showed MRSA induced abscess in wistar rat. (a) showed MRSA induced abscess in wistar rat. (b) showed MRSA induced abscess isolated from rat.

3.4. Lipid profile

The LZD treated infected rats showed (Table 2) significant ($p < 0.05$) elevation in the cholesterol and triglycerides in comparison to the normal group. SN and LZD treated rats showed no significant ($p > 0.05$) changes in lipid levels when compared to normal.

3.5. Enzymatic biomarkers

The LZD treated infected rats showed (Fig. 3) significant ($p < 0.01$) elevation in the AST, ALT, ALP, LDH levels in comparison to the normal group. It indicates LZD treatment induce liver damage. LZD and SN-treated groups showed the minimum elevation of AST, ALT, ALP and LDH levels. The LZD treated infected animals showed 2.19 mU/mg of IAP. SN 25, 50, and 100 mg/kg administration along with LZD treated animals showed 3.39 , 3.76 , and 4.08 mU/mg of IAP levels respectively represented in Table 3.

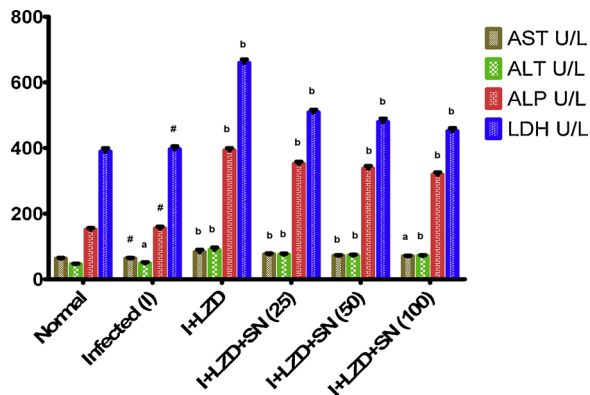
3.6. Serum biochemical parameters

LZD treated infected animals showed (Table 3) significant ($p < 0.01$) elevation (20.1 mg/dl) of lactic acid and bilirubin indicates liver damage, but no significant change in urea and creatinine when compared to control. The elevated lactic acid is the marker for mitochondrial toxicity. LZD and SN (25 , 50 and 100 mg/kg) treated groups showed decreased (18.6 , 16.5 , and 15.7 mg/dl) lactic acid. SN has dose-dependent protective activity against mitochondrial toxicity induced by LZD.

Table 2

Effect of silymarin on hematological parameters in control and LZD treated MRSA infected rats.

Group	RBC ($\times 10^6/\mu\text{l}$)	WBC ($\times 10^3/\mu\text{l}$)	PLT ($\times 10^5/\mu\text{l}$)	Hb (gm/dl)	CH (mg/dl)	TG (mg/dl)
Normal	6.24 $\pm$ 0.2	12.38 $\pm$ 0.6	8.28 $\pm$ 0.1	12.18 $\pm$ 0.3	82.3 $\pm$ 1.2	68.9 $\pm$ 1.6
Infected	6.14 $\pm$ 0.3 [#]	13.02 $\pm$ 0.1 ^b	7.94 $\pm$ 0.3 [#]	12.10 $\pm$ 0.2 [#]	81.1 $\pm$ 3.4 [#]	68.1 $\pm$ 1.2 [#]
I + LZD	4.93 $\pm$ 0.1 ^b	10.32 $\pm$ 0.2 ^b	4.96 $\pm$ 0.2 ^b	10.03 $\pm$ 0.1 ^b	91.9 $\pm$ 1.4 ^a	89.9 $\pm$ 1.5 ^a
I + LZD + SN (25)	5.41 $\pm$ 0.2 ^a	10.94 $\pm$ 0.3 ^b	5.06 $\pm$ 0.4 ^b	10.52 $\pm$ 0.2 ^b	88.2 $\pm$ 1.6 [#]	81.4 $\pm$ 2.1 [#]
I + LZD + SN (50)	5.92 $\pm$ 0.3 [#]	11.03 $\pm$ 0.2 ^b	5.26 $\pm$ 0.1 ^b	10.88 $\pm$ 0.3 ^b	86.1 $\pm$ 2.4 [#]	76.5 $\pm$ 2.2 [#]
I + LZD + SN (100)	5.98 $\pm$ 0.1 [#]	11.18 $\pm$ 0.4 ^a	5.39 $\pm$ 0.2 ^b	10.93 $\pm$ 0.1 ^b	84.6 $\pm$ 1.2 [#]	75.8 $\pm$ 2.5 [#]

Values are mean $\pm$ S.D; n = 10 in each group; Data were analyzed by one way ANOVA followed by Dunnett's test.[#] No significant difference in comparison with the normal group (P > 0.05).^a Significant difference in comparison with the normal group (P < 0.05).^b Significant difference in comparison with the normal group (P < 0.01).**Fig. 3.** Effect of silymarin treatment on liver marker enzymes in LZD induced hepatotoxicity different experimental groups. Values are mean $\pm$ S.D; n = 10 in each group; Data were analyzed by one way ANOVA followed by Dunnett's test.[#]No significant difference in comparison with the normal group (P > 0.05).^aSignificant difference in comparison with the normal group (P < 0.05).^bSignificant difference in comparison with the normal group (P < 0.01).

3.7. Histopathological results

3.7.1. Histopathology of liver

Histopathological findings of liver were shown in Fig. 4. Microscopic examination of control and infected rats showed normal hepatic architecture (Fig. 4a and b). The LZD treated infected rats revealed grade 3 macrovesicular steatosis (Fig. 4c showed two-thirds of hepatocytes contained microvesicles of fat), apoptotic hepatocytes with diffuse vacuolization of hepatocytes and grade 2 inflammation. The SN (25 mg/kg) and LZD treated rats showed bile duct hyperplasia, grade 2 macrovesicular steatosis, and inflammation (Fig. 4d). SN (50 and 100 mg/kg) and LZD treated rats showed regenerative hepatocytes grade 1 macrovesicular steatosis and inflammation (mild infiltration of

mononuclear cells) and (Fig. 4e and f). (Grading: Macrovesicular steatosis: grade 0 absent; grade 1 < 33%; grade 2: 33–66%; grade 3 > 66%. Inflammation: 0 normal; 1 mild; 2 moderate; 3 severe). The total score was calculated (Table 4) by averaging all the scores of each group.

3.7.2. Histopathology of kidney

Histopathological findings of kidney were shown in Fig. 5. The LZD treated rats showed mild blood vessel congestion in tubular cells but no necrotic cells (Fig. 5c). The SN 100 mg/kg and LZD-treated rats showed few hyaline casts (Fig. 5f). The control, infected, SN 25 and SN 50 mg/kg treated groups (Fig. 5a, b, d, and e) showed normal glomerular and tubular cells.

3.7.3. Histopathology of bone marrow

Bone marrow smear of LZD-treated rat showed a reduction in the cellular portion of bone marrow (Fig. 6c) and hypoplasia of hemopoietic cells. SN (25 & 50 mg/kg) and LZD treatment (Fig. 6d and e) showed improvement in hemopoietic cell size. Bone marrow smear of SN 100 mg/kg and LZD treated rats showed (Fig. 6f) normal cell size and mild improvement in cellular density than LZD treated an animal. Fig. 6a and b control and infected animals revealed normal bone marrow density.

3.8. Lipid peroxidation

LZD treated infected animals showed a significant (P < 0.01) decrease in the level of total protein and an increase in the level of lipid peroxidation markers malondialdehyde (MDA) and lipid hydroperoxides (LH) when compared to normal control. Treatment with SN and LZD showed a significant increase in protein and a decrease in MDA and LH (Table 5).

Table 3

Effect of silymarin on biochemical parameters in control and LZD treated MRSA infected rats.

Group	Lactic acid mg/dl	Urea mg/dl	Creatinine mg/dl	Bilirubin mg/dl	IAP mU/mg of tissue
Normal	12.6 $\pm$ 0.4	26.6 $\pm$ 0.5	0.76 $\pm$ 0.06	0.67 $\pm$ 0.02	4.49 $\pm$ 0.08
Infected (I)	12.8 $\pm$ 0.2 [#]	24.8 $\pm$ 0.6 [#]	0.77 $\pm$ 0.06 [#]	0.68 $\pm$ 0.03 [#]	4.54 $\pm$ 0.07 [#]
I + LZD	21.6 $\pm$ 0.6 ^b	27.6 $\pm$ 0.3 [#]	0.76 $\pm$ 0.05 [#]	0.96 $\pm$ 0.04 ^b	2.19 $\pm$ 0.06 ^b
I + LZD + SN (25)	18.6 $\pm$ 0.4 ^b	27.6 $\pm$ 0.1 [#]	0.75 $\pm$ 0.03 [#]	0.91 $\pm$ 0.03 ^b	3.39 $\pm$ 0.02 ^b
I + LZD + SN (50)	16.5 $\pm$ 0.2 ^b	27.4 $\pm$ 0.2 [#]	0.76 $\pm$ 0.04 [#]	0.82 $\pm$ 0.02 ^b	3.76 $\pm$ 0.04 ^b
I + LZD + SN (100)	15.7 $\pm$ 0.1 ^b	27.1 $\pm$ 0.3 [#]	0.74 $\pm$ 0.02 [#]	0.73 $\pm$ 0.01 ^a	4.08 $\pm$ 0.06 ^a

Values are mean $\pm$ S.D; n = 10 in each group; Data were analyzed by one way ANOVA followed by Dunnett's test.[#] No significant difference in comparison with the normal group (P > 0.05).^a Significant difference in comparison with the normal group (P < 0.05).^b Significant difference in comparison with the normal group (P < 0.01).

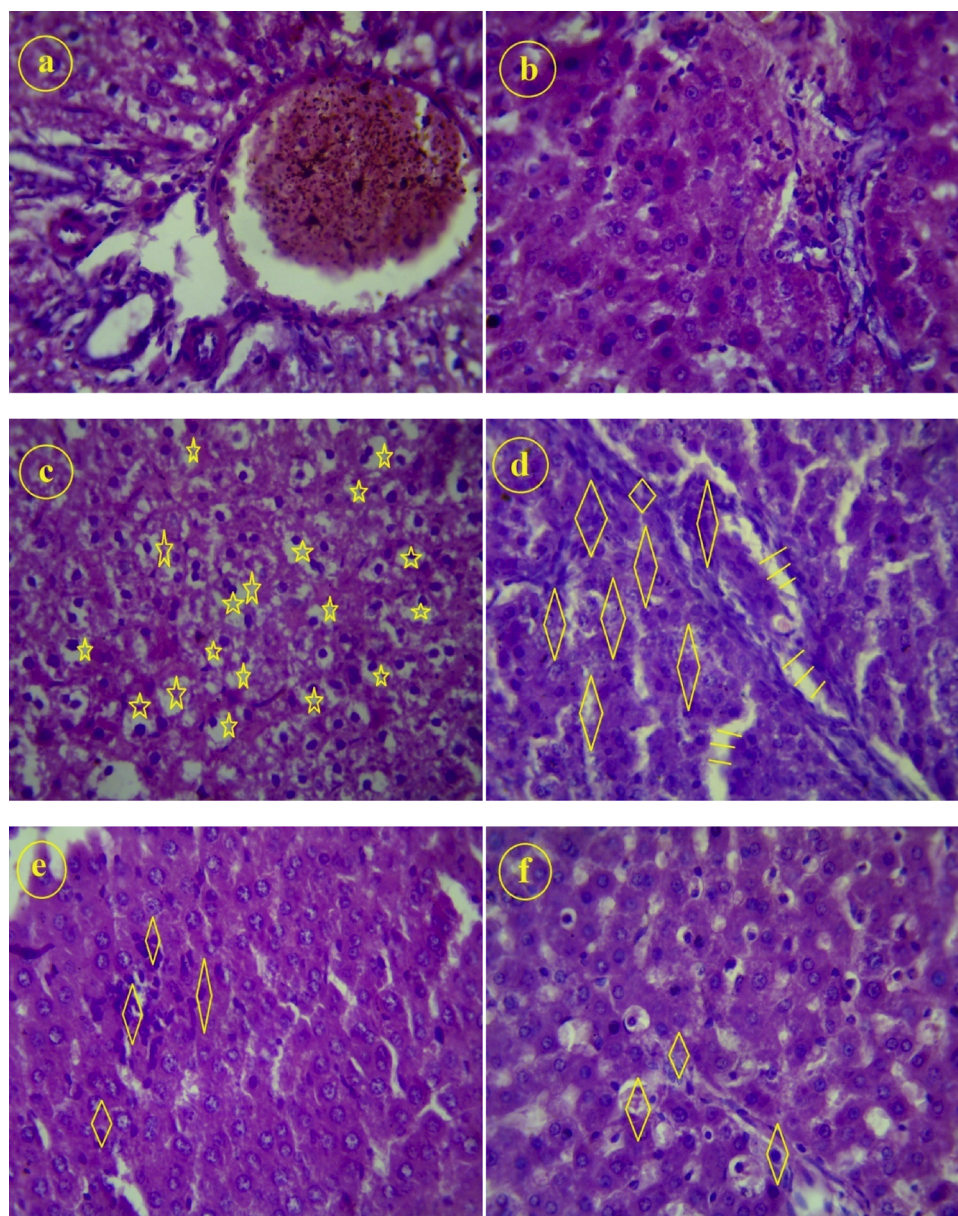


Fig. 4. Histopathology of Liver. (a–f) represents the histopathology of LZD and SN treated liver under 40X magnification. Fig. 4a and b control and infected rats showed the normal liver architecture. Fig. 4c LZD treated infected rats showed macrovesicular steatosis (star). Fig. 4d SN (25) treated rats showed moderate inflammation (rectangular) and bile duct hyperplasia (line) in LZD treated infected rats. Fig. 4e and f revealed SN (50 and 100) treated rats showed mild inflammation (rectangular) and regenerative changes in LZD treated infected rats.

Table 4

Effect of silymarin on histopathological scoring of macrovesicular steatosis and inflammation of hepatocytes in control and LZD treated MRSA infected rats.

Group	Hepatocytes scoring	
	Macrovesicular steatosis	Inflammation
Normal	0.0	0.0
I + LZD	2.8 ± 0.42 ^a	2.2 ± 0.42 ^a
I + LZD + SN (25)	1.7 ± 0.48 ^b	1.4 ± 0.51 ^b
I + LZD + SN (50)	1.2 ± 0.42 ^b	0.9 ± 0.31 ^b
I + LZD + SN(100)	0.9 ± 0.31 ^b	0.7 ± 0.48 ^b

Values are mean ± S.D; n = 10 in each group; Data were analyzed by one way ANOVA followed by Dunnett's test.

^a Significant difference in comparison with the normal group (P < 0.01).

^b Significant difference in comparison with the LZD treated group (P < 0.01).

3.9. Oxidative stress biomarkers in liver

The changes in the levels of enzymatic antioxidants SOD, CAT, GSSH, GPx and Px in the liver of control and experimental animals are shown in Table 5. A significant decrease in the levels of these radical scavengers was noted after 14 days treatment with LZD. Upon administration of SN for 14 days, the activities of enzymatic antioxidants were significantly altered as compared to LZD treated rats.

3.10. Non- enzymatic stress biomarkers in liver

LZD treated infected rats showed (Fig. 7) significant reduction in GSH, Vitamin C, A, and E when compared to control. Treatment with the SN significantly (P < 0.01) restored the levels of GSH, Vitamin C, A and Vitamin E levels which is almost similar to the control group.

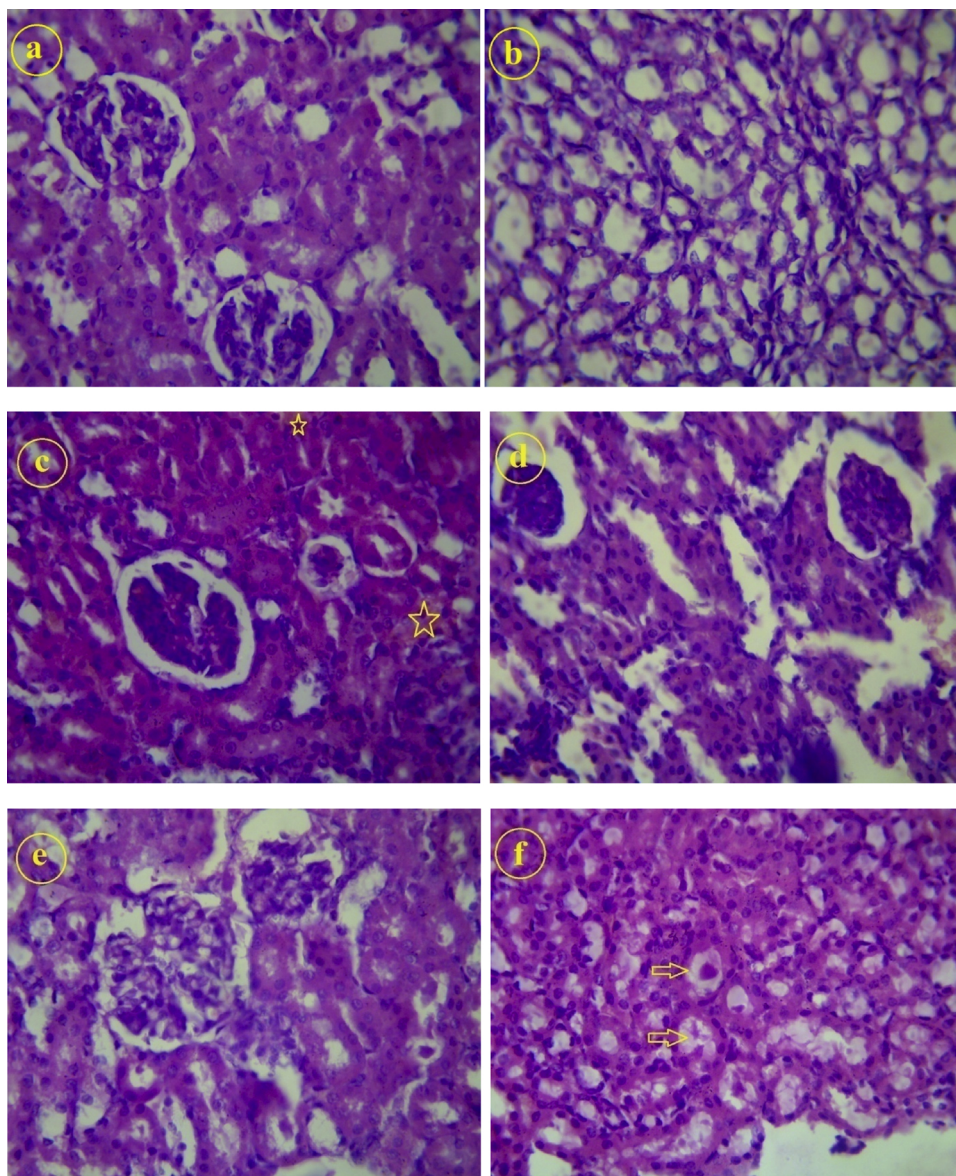


Fig. 5. Histopathology of Kidney (a–f) represents the histopathology of LZD and SN treated kidney under 40X magnification. The control and infected rats displayed the normal kidney architecture. Fig. 5a showed complete Bowman's capsule and glomerular architecture. Fig. 5b showed normal tubular epithelium. Fig. 5c LZD treated infected rats showed mild blood vessel congestion (star). Fig. 5d and e SN (25 and 50) treated rats showed normal glomerular cells. Fig. 5f SN100 treated rats revealed one focal showed hyaline cast (arrow) in LZD treated infected rats. Histopathology of Bone marrow.

4. Discussion

The present study was designed to investigate the hepatoprotective and antioxidant potential of SN against LZD treated MRSA infected Wistar rats. MRSA may cause invading or life-threatening infections because it is resistant to commonly used several antibiotics and inappropriate to handle. Skin and soft tissue infections (SSTI) caused by *Staphylococcus*, *Enterococcus*, *Pseudomonas*, and *Streptococcus*. Multidrug-resistant, gram-positive pathogen MRSA, is a predominant pathogen for SSTI. The antibiotics available to treat MRSA infections are limited. Linezolid, tigecycline, clindamycin, vancomycin, and daptomycin showed efficacy, safety, clinical cure and eradication rates in MRSA-caused SSTI [30]. The LZD has superior to other antibiotics in the treatment of MRSA infections. The antimicrobial oxazolidinones, discoverer DuPont in the 1980s investigated early derivatives [31], and newer potent molecules (PNU-140693 and PNU-141059) [32] are induce hepatotoxicity in animal models. The previous clinical studies also reported that LZD therapy causes hepatotoxicity and other serious side

effects such as lactic acidosis, and bone marrow suppression [33].

In this study, LZD administration leads to a significant elevation in serum AST, ALT, and ALP which is a most sensitive marker in the diagnosis of liver damage. Furthermore elevation in serum LDH, bilirubin, and decreased liver protein was observed due to the release of these enzymes and protein into the circulation after hepatocellular damage. Further, apoptotic hepatocyte was observed in a histological report suggesting, LZD induce liver damage in rats. Oral treatment of SN attenuated LZD induced elevation in the hepatic enzymes, improvement in liver protein level and cytological findings associated with LZD induced hepatotoxicity by conserving the membrane integrity of cellular organelles and stimulation of protein synthesis in the liver [34].

Several kinds of researches indicate that free radicals play a main role in numerous pathological dysfunctions such as hepatotoxicity, cardiovascular disorders, inflammations, and diabetes mellitus [35,36]. Lipid peroxidation is initiated by free radicals, its damage cell membrane of hepatocytes and other peripheral tissues and causes a wide

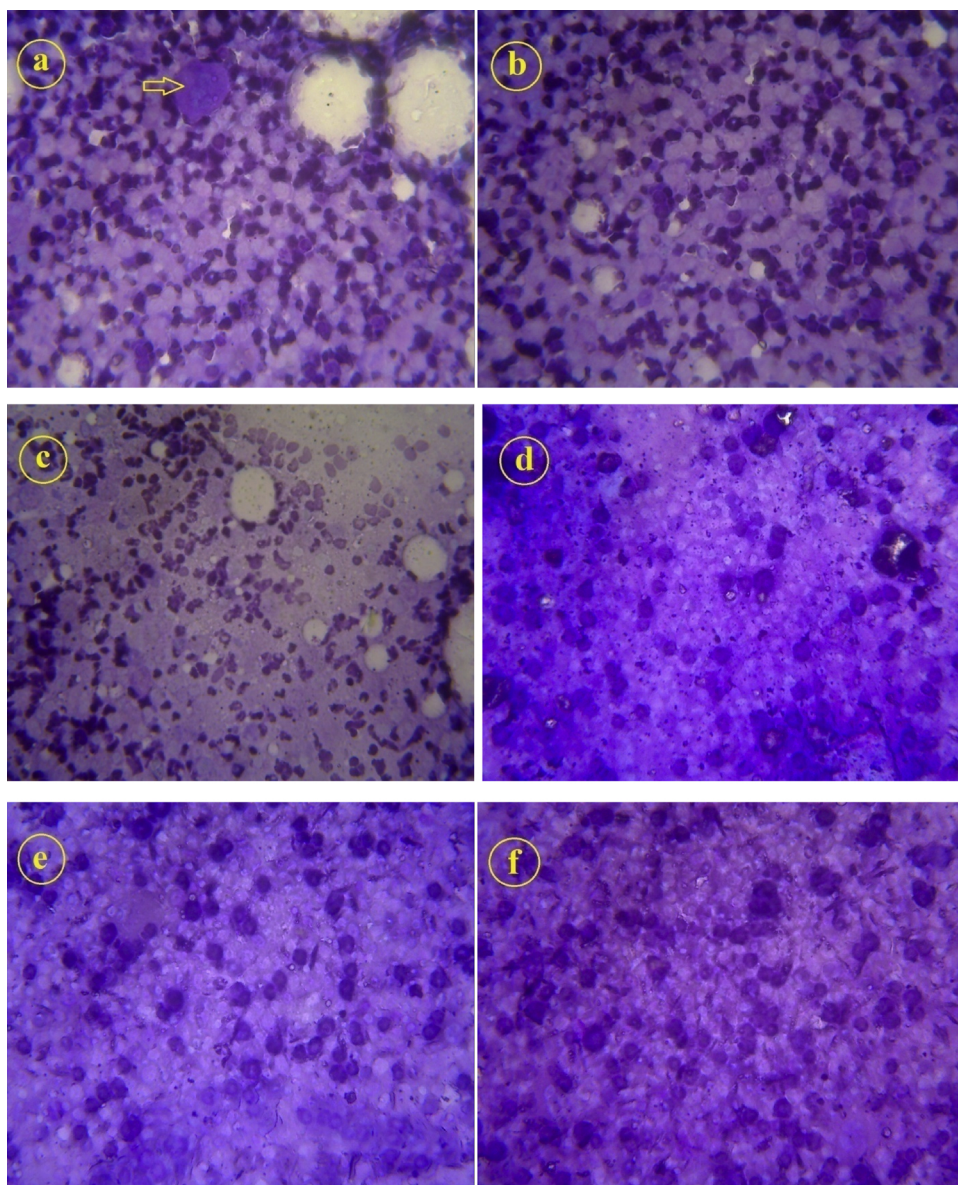


Fig. 6. (a–f) represents the bone marrow smear of LZD and SN treated groups under 40X magnification. Fig. 6a control animal showed normal cellular density and one megakaryocyte (arrow). Fig. 6b infected rats revealed normal bone marrow density. Fig. 6c Bone marrow smear of LZD treated rat showed hypocellularity and hypoplasia of hemopoietic cells. Fig. 6d SN 25 mg/kg & LZD treated animal showed mild improvement in cellular density. Fig. 6e SN 50 mg/kg and LZD treated rats showed enhancement in hemopoietic cell size. Fig. 6f Bone marrow smear of SN 100 mg/kg and LZD treated rats showed mild improvement in cellular density and normal size of hemopoietic cells.

spectrum of damages [37]. In the current study, LZD treated rats showed a significant increase in CH, TG in blood and increased fat accumulation leading to microvesicular steatosis in liver histological architecture. Further, LZD treated rats showed a significant increase in lipid peroxidation markers such as MDA, LH in the liver and cause free

radical-induced liver damage. SN administration significantly decreases the circulatory lipid levels and fat accumulation in the liver cells. This finding was consistent with Muriel and Mourelle [38] Mourelle and Franco [39] who demonstrated SN is associated with changes in CH and lipid levels in the liver and other compartments. SN treatment was

Table 5

Effect of silymarin on liver protein, lipid peroxidation markers and enzymatic antioxidants levels in control and LZD treated MRSA infected rats.

Group	Protein (mmole/g wet tissue)	MDA (nmoles/mg protein)	LH (nmoles/mg protein)	CAT (μmoles/ mg protein)	SOD (nmoles/ mg protein)	GSSH (nmoles/ mg protein)	Px (nmoles/ mg protein)	GPx (nmoles/ mg protein)
Normal	191.4 ± 6.2	0.78 ± 0.02	0.62 ± 0.01	42.4 ± 2.1	5.18 ± 0.02	32.4 ± 2.6	8.68 ± 0.01	9.26 ± 0.03
Infected (I)	184.8 ± 6.3 [#]	0.91 ± 0.03 ^b	0.68 ± 0.04 [#]	40.1 ± 1.8 [#]	4.95 ± 0.02 [#]	31.9 ± 2.1 [#]	8.72 ± 0.02 [#]	9.04 ± 0.02 [#]
I + LZD	152.1 ± 8.4 ^b	0.96 ± 0.02 ^b	0.84 ± 0.02 ^b	39.1 ± 1.2 [#]	4.67 ± 0.03 ^b	28.2 ± 1.2 ^b	8.26 ± 0.04 ^b	8.14 ± 0.01 ^b
I + LZD + SN (25)	158.1 ± 6.2 ^a	0.92 ± 0.02 ^b	0.80 ± 0.01 ^b	39.6 ± 1.8 [#]	4.74 ± 0.02 ^a	28.9 ± 1.2 [#]	8.35 ± 0.01 ^b	8.37 ± 0.02 ^b
I + LZD + SN (50)	172.2 ± 8.6 ^a	0.87 ± 0.01 ^b	0.78 ± 0.02 ^b	40.8 ± 1.4 [#]	4.81 ± 0.01 ^a	29.7 ± 1.8 [#]	8.42 ± 0.02 ^b	8.69 ± 0.03 ^b
I + LZD + SN (100)	175.7 ± 5.4 [#]	0.83 ± 0.02 ^b	0.76 ± 0.01 ^b	41.2 ± 2.5 [#]	4.94 ± 0.01 [#]	30.4 ± 2.1 [#]	8.49 ± 0.01 [#]	8.92 ± 0.02 ^a

Values are mean ± S.D; n = 10 in each group; Data were analyzed by one way ANOVA followed by Dunnett's test.

[#] No significant difference in comparison with the normal group (P > 0.05).

^a Significant difference in comparison with the normal group (P < 0.05).

^b Significant difference in comparison with the normal group (P < 0.01).

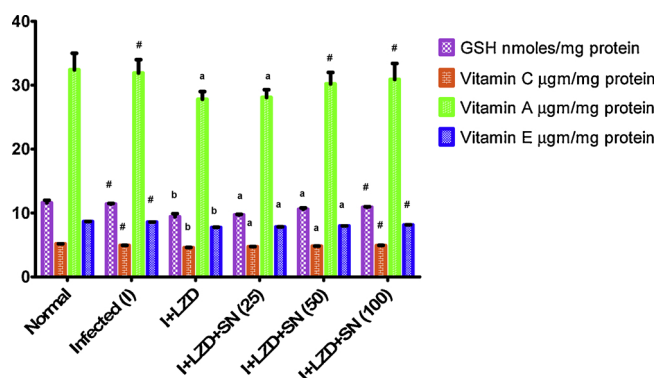


Fig. 7. Effect of silymarin on non-enzymatic antioxidant levels in LZD treated MRSA infected rats. Values are mean $\pm$ S.D; n = 10 in each group; Data were analyzed by one way ANOVA followed by Dunnett's test.

No significant difference in comparison with the normal group ($P > 0.05$).

a Significant difference in comparison with the normal group ($P < 0.05$).

b Significant difference in comparison with the normal group ($P < 0.01$).

significantly exhibited anti lipoperoxidation effect by inhibiting the free radical formation and effectively prevents liver damage.

SOD is the first enzyme in the antioxidant defense mechanism. It scavenges superoxide anion and converts it into hydrogen peroxide. Catalase protects the cell from oxidative damage induced by H_2O_2 and hydroxyl radical. The enzyme GSSH catalyzes the conversion of oxidized glutathione (GSSG) to reduced glutathione (GSH) and maintaining the normal GSH level. SN treated rats significantly alters antioxidant enzymes include SOD, CAT, GPX, PX, GSSH, and these enzymes are the first line of defense against reactive oxygen species. SN showed free scavenging activity, which is a beneficial action against pathological alteration caused by free radicals and restored the enzymatic antioxidants near to normal [40].

Reduced glutathione inhibits oxidative stress, decrease lipid peroxidation and synergistic with vitamin-A and E. Vitamin-C also scavenge the free radicals in combination with GSH and vitamin-E. It prevents the formation of excessive free radicals by regenerating the reduced form of vitamin-E [41]. The decreased levels of these antioxidant vitamins and GSH observed in LZD treatment. SN treatment protects against the free radical-induced damage by restored non-enzymatic antioxidant levels.

The intestinal microbiota is a pivotal role in maintaining human health and well being. Antibiotics cause dysbiosis, an imbalance in the number and composition of intestinal bacteria, which facilitate the growth of opportunistic bacteria. Gut enzyme IAP physiologically, exhibit many functions, including decreasing dietary fat absorption and detoxifying bacterial toxins [42]. In the current study, the LZD treated rats showed a significant reduction in IAP cause alterations in fat absorption obviously increased serum lipid levels, accumulation of fat and fatty degeneration in liver was observed. SN administration along with LZD showed a significant increase in the IAP level and decreases the fatty degeneration in hepatocytes evidenced by the absence of macrovesicular steatosis in histopathological results. SN administration attenuated LZD induced an increase in circulatory CH and TG levels by inhibiting HMG-CoA reductase enzyme [43].

LZD inhibits bacterial protein synthesis by binding residues in the 23S ribosomal RNA of the 50S bacterial ribosomes. LZD binds to human ribosomes, leading to a decrease in the activity of respiratory chain complexes having mitochondrial DNA-encoded subunits and decrease the protein of the complexes, due to similarities between bacterial and human ribosomes [44]. The LZD treatment decreased the mitochondrial enzyme activity in tissues. LZD is metabolized by oxidation of its morpholine ring in the liver with a utilization of NADPH [45] and more than 60% of lactic acid is also metabolized by the liver. Therefore, liver dysfunction can be considered as a risk factor for LZD-induced lactic

acidosis. The LZD induced lactic acidosis due to mitochondrial DNA polymorphisms [46]. The deposition of lactic acid induced by LZD also damages the organs such as the liver and kidney. The present study indicates LZD treated rats showed liver damage and normal histological architecture of kidney evidenced by normal serum urea and creatinine. The SN intake promotes the lactate uptake by the liver and decreases the circulating lactate level. The high blood lactate is cleared 70% by the liver, 10–20% with kidneys, and low degree with skeletal muscles [47]. SN has increasing lactic acid clearance and prevent lactate deposition in organs was confirmed by decreasing of serum lactate level in dose depending manner of SN administration. The SN treated rats showed mild inflammation and regeneration of hepatocytes. The SN 25, and 50 mg/kg treated rats showed normal kidney cells but SN 100 mg/kg treated rat's histopathology of kidney cells showed only one focal area of the hyaline cast may indicate excretion of high acidic urine as an evidence of excretion of lactic acid and the decreased serum lactate level.

Several studies reported that the prolonged treatment of LZD showed reversible bone marrow suppression and red cell hypoplasia characterized by anemia and thrombocytopenia. [48–50]. The mechanism of this effect is not known, but LZD induced myelosuppression may result from inhibition of mitochondrial protein synthesis by inhibiting [^{35}S] methionine incorporation into mitochondrial proteins [51]. The present study indicates the LZD-induced toxic effects on blood parameters was demonstrated by declined in RBC, WBC, PLT, and Hb. The LZD-induced myelosuppression was noticeably by hypocellularity and hypoplasia of hemopoietic cells in bone marrow smear. LZD treatment showed anemia due to suppressing hemopoietic cells, erythroid precursor cells sensitive to mitochondrial protein synthesis inhibition [52], loss of mitochondrial function in hematopoietic cells [53], and increased lipid peroxidation causes reactive oxygen species induced hemolysis [54]. Oral administration of SN restored the RBC level, mild improvement of WBC, PLT, and Hb towards normal. Bone marrow smear of SN treated rats showed mild improvement in cellular density and restored the normal size of hemopoietic cells. These results are in agreement with other studies which reported SN protects hemolysis induced by oxidative stress because of its anti-oxidant and anti-inflammatory action [6,55]. The LZD-treated groups showed significant hematological adverse effects on the 15th day of the experiment and SN administration had successfully protected these hematological disturbances.

Additionally, SN use along with LZD enhanced the antibacterial activity against MRSA infection. It was confirmed by the SN and LZD treated rats showed the reduction in abscess size, weight, and bacterial load. The phenolic compounds of SN bind to extracellular proteins and form a complex that binds to the bacterial cell wall. It alters the membrane permeability and favors the penetration of antibiotics [56].

5. Conclusion

Based on the study, it was demonstrated that the concomitant oral administration of SN has proven to reduce LZD-induced hepatotoxicity and oxidative stress in a dose-dependent manner against MRSA infected rats. Moreover, SN treatment exhibited a significant ameliorative potential by attenuated the myelosuppression and liver damage provoked by lactic acidosis, IAP in LZD-treated rats. The present study indicated that the on SN also reinforced with numerous prospectives in the management of hepatotoxicity. These outcomes are of clinical importance particularly, use of SN as a safe unique therapy or in combination to counteract LZD-induced hepatotoxicity. The synergistic antibacterial role of SN in combination with LZD is well supported; further study needed for the different diseased population while using antibiotic LZD.

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Conflict of interest

The authors have no conflict of interest with this project.

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